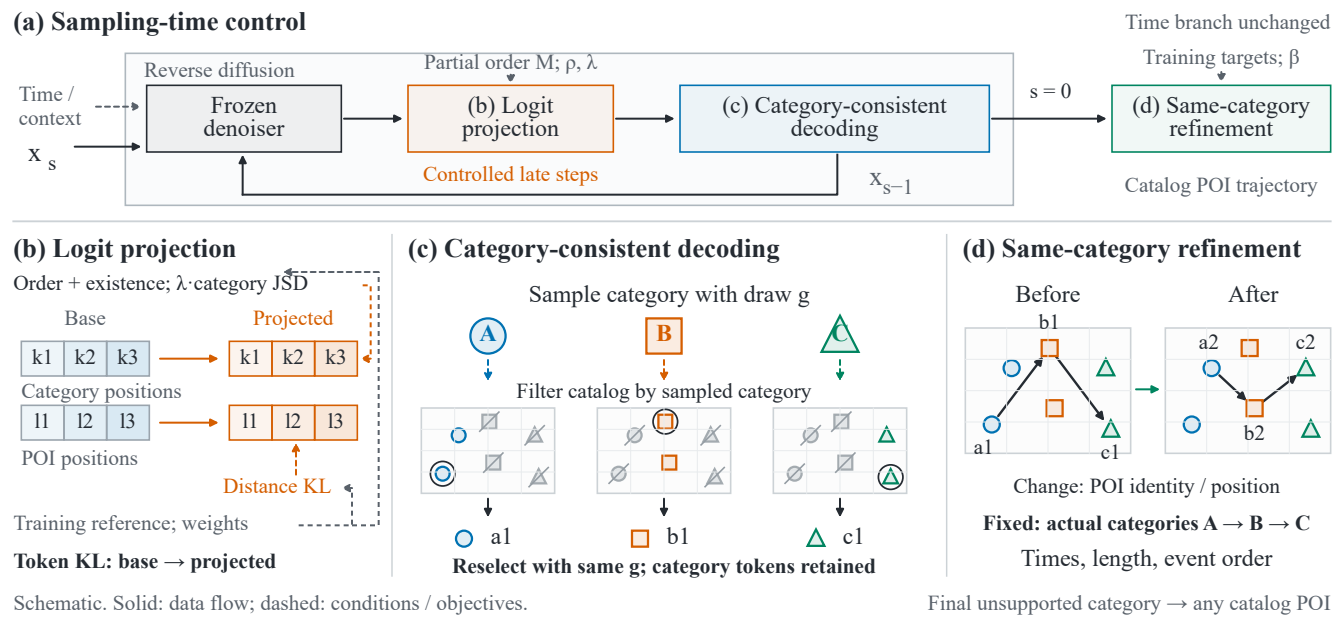

(a) Sampling-time control
Time branch unchanged
Training targets; β
Reverse diffusion
Partial order M; ρ, λ
Time / context
x s
Frozen denoiser
(b) Logit projection
(c) Category-consistent decoding
s = 0
(d) Same-category refinement
Controlled late steps
x s−1
Catalog POI trajectory
(b) Logit projection
Order + existence; λ·category JSD
Base
Projected
k1 k2 k3
Category positions
l1 l2 l3
POI positions
Distance KL
Training reference; weights
Token KL: base → projected
(c) Category-consistent decoding
Sample category with draw g
A
B
C
Filter catalog by sampled category
a1
b1
c1
Reselect with same g; category tokens retained
(d) Same-category refinement
Before
After
b1
a2
c2
a1
c1
b2
Change: POI identity / position
Fixed: actual categories A → B → C
Times, length, event order
Schematic. Solid: data flow; dashed: conditions / objectives.
Final unsupported category → any catalog POI